Rose, sunflower or tulip?

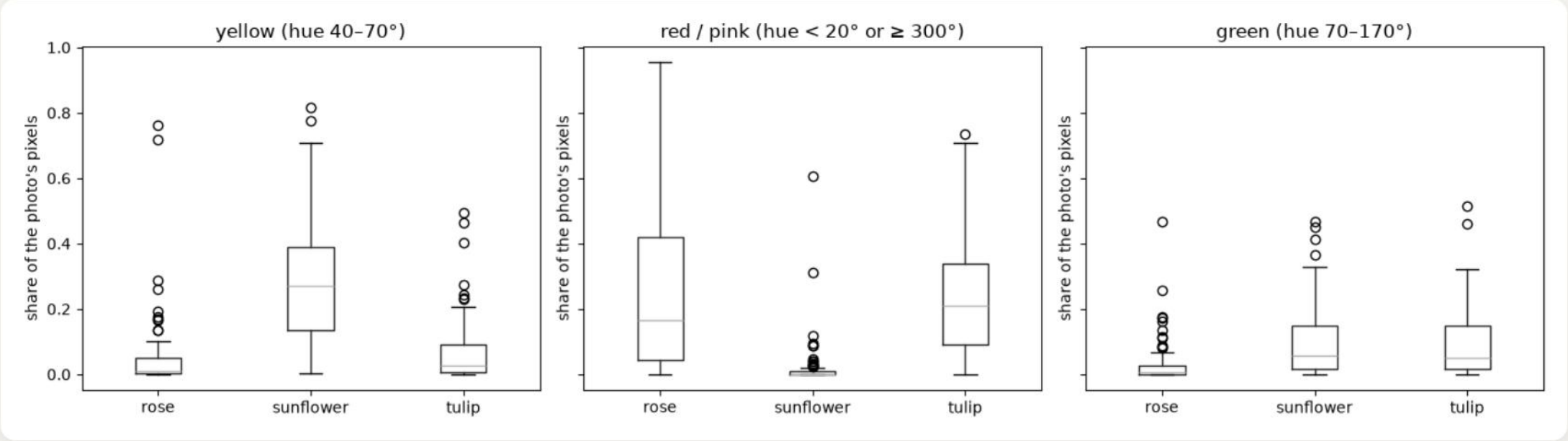


- A garden centre sends a care guide for a customer's flower photo
- 390 photos, only 100 per flower to learn from
- And: which photos does the model get wrong?

Rules fixed before the first model

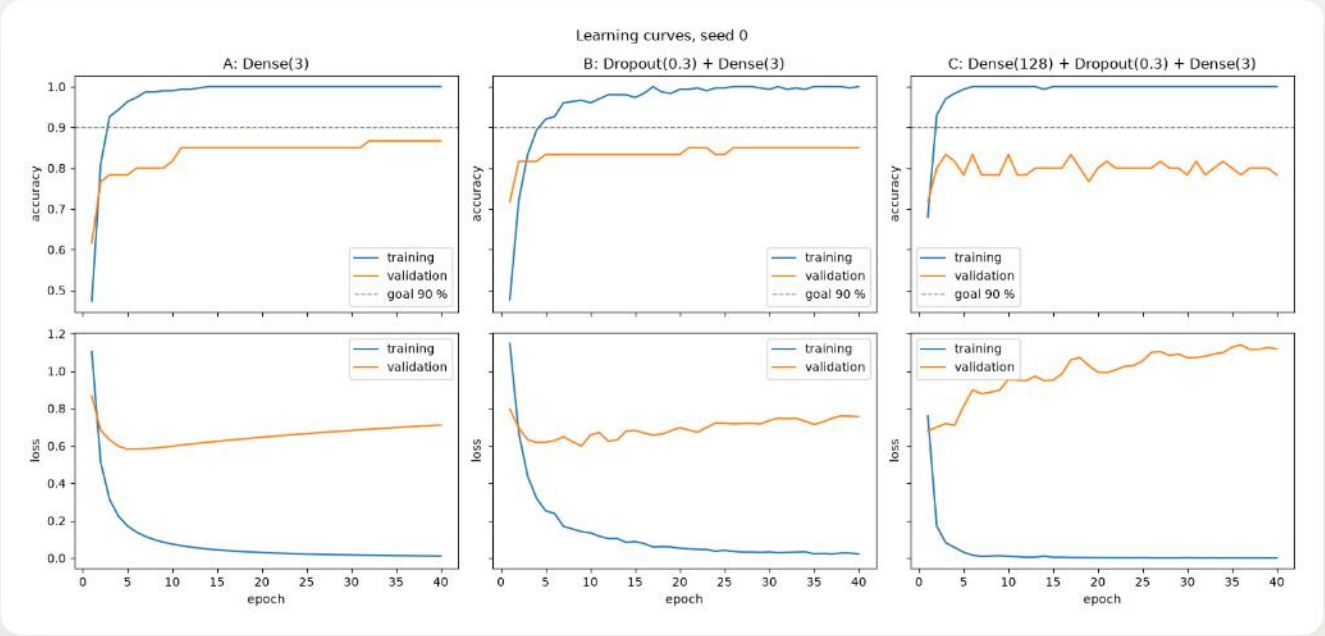
- Metric: accuracy, the share of photos with the right care guide
- Goal: at least 90 % on the validation photos = at most 6 of 60 wrong
- Every photo squeezed to 160 × 160 pixels, exactly like the app
- The 30 test photos stay locked until the very end
- Extras: a staff-check threshold and a nicer app

Colour or shape?



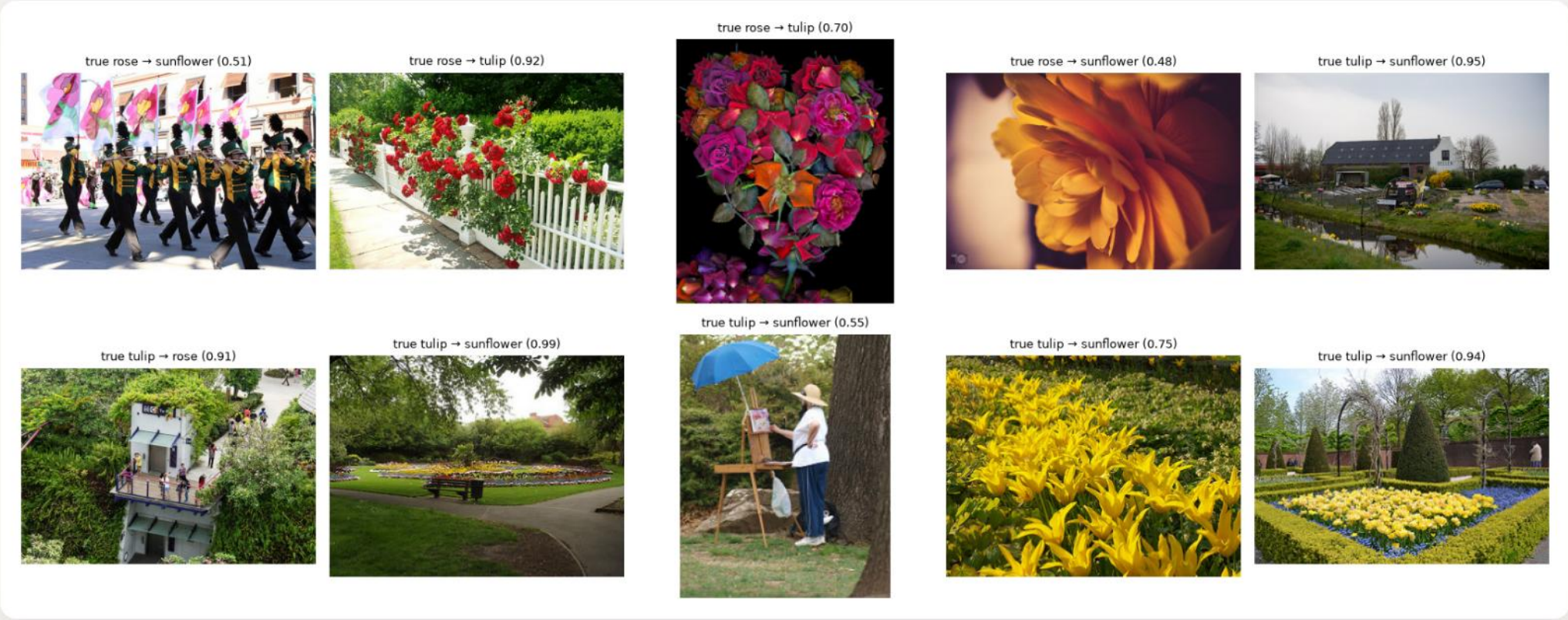
- Our guess: colour. Half right
- Roses 17 %, tulips 21 % red or pink: the shape has to decide
- Sunflowers: 27 % of the pixels bright yellow
- Validation independent: 6 of 60 shared photographers, no copies

A pretrained network, frozen



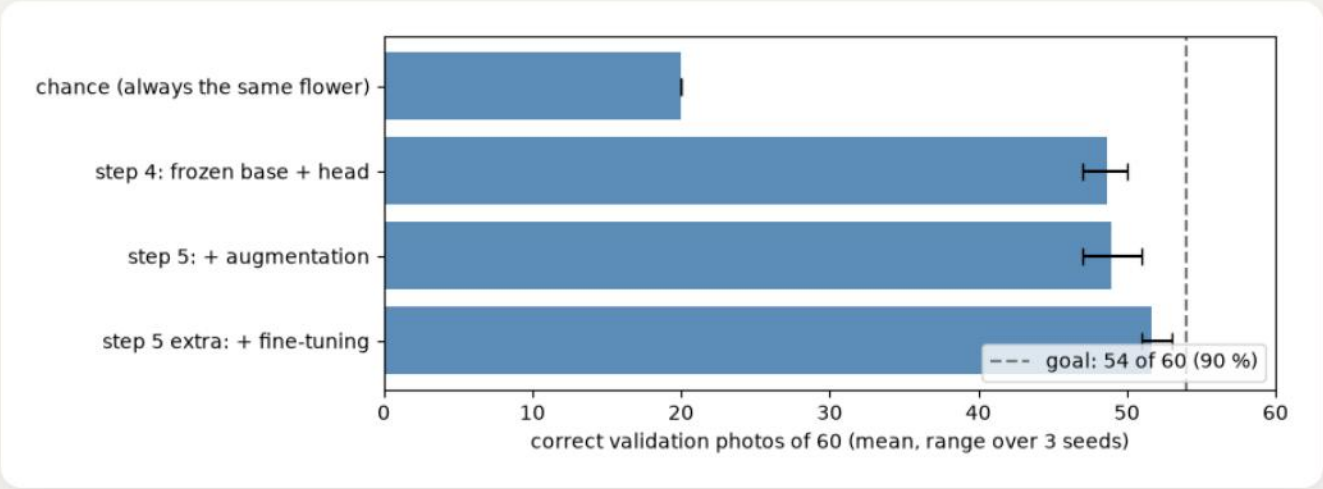
- MobileNetV2, trained on 1.2 million ImageNet photos
- 2,261,827 parameters, only 3,843 trainable
- Our guess: 85 %. Result: 84 %
- Overfitting after about 5 epochs

Where it fails: small flowers in big scenes



- Tulip → sunflower: 5 of 10 errors
- Small yellow tulips in parks
- Up to 0.99 sure

Augmentation and fine-tuning



- Augmentation: 49.0 vs 48.7 of 60. No extra hits
- … but better probabilities and later overfitting
- Fine-tuning: 51.7 of 60 on average = 86 %
- Goal 54 of 60: missed, and not lowered

When should a person check? Threshold 0.90

44 of 60

answered alone above 0.90

40

of them right (91 %)

16

go to a member of staff

4

confident mistakes slip through

Fixed on the validation photos. A threshold cannot catch a model that is sure and wrong.

One test, opened once

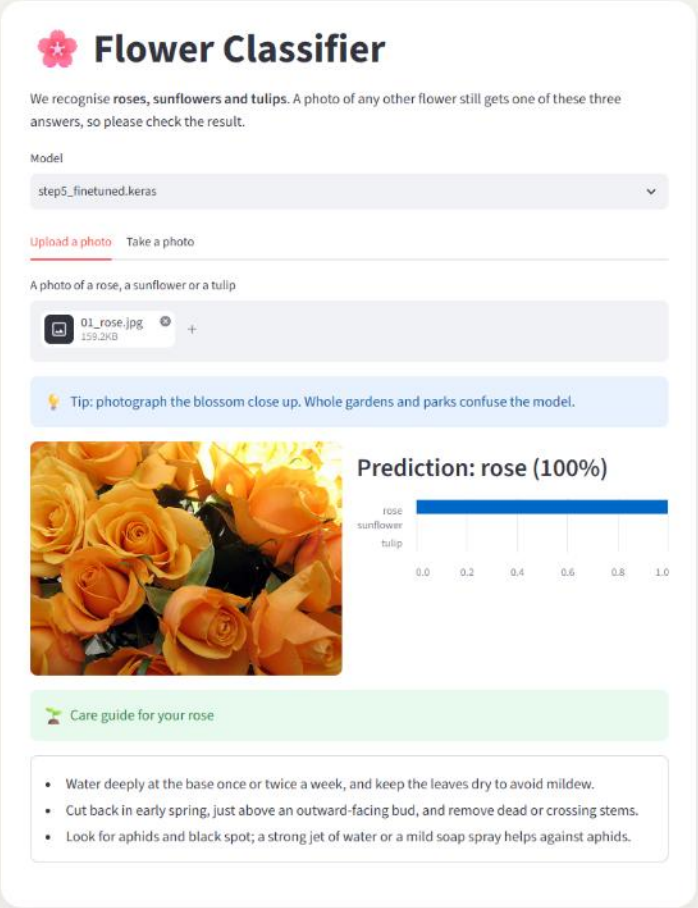
29 of 30 right = 96.7 %

- 95 % interval: 82.8 % to 99.9 %. Our validation estimate (86 %) fits
- The one error: sunflowers in a vegetable garden → rose, only 0.66 → staff
- With the threshold: 26 of 30 answered alone, all right

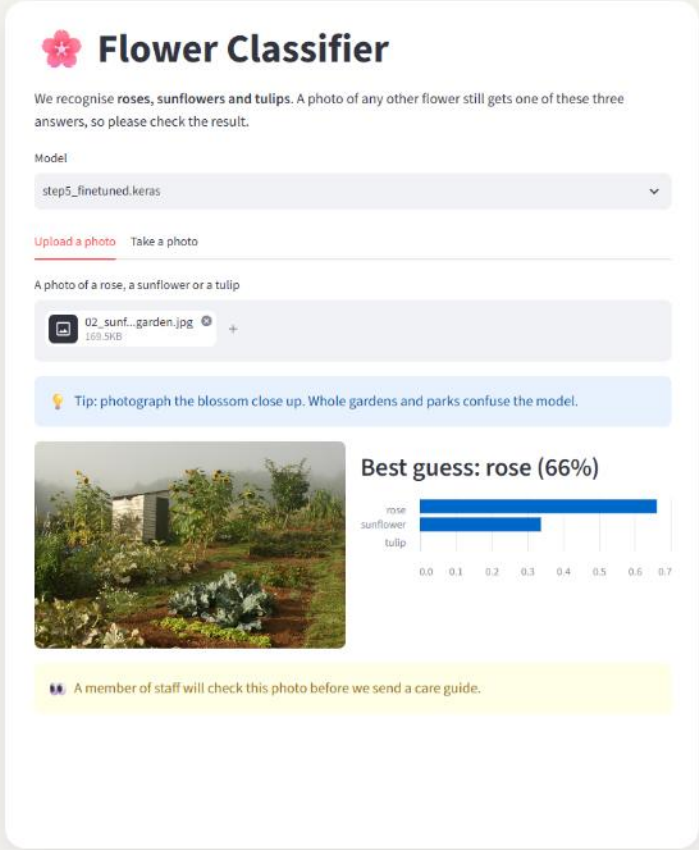


true: sunflower · predicted: rose (0.66)

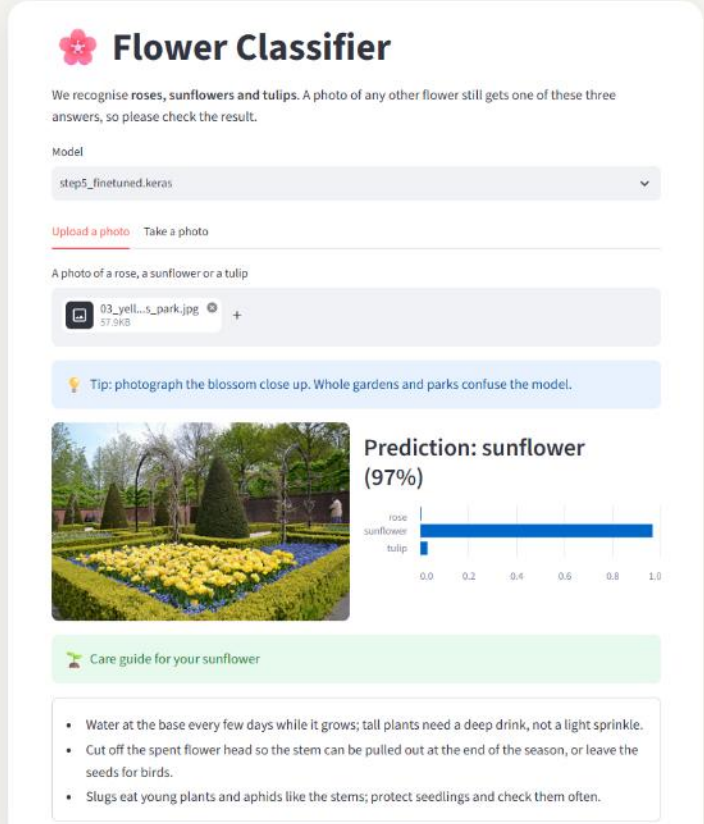
The app: care guide, staff check, close-up tip



Clear photo: rose 100 %, care guide



Garden: best guess 66 %, staff check



Known weakness: yellow tulips → sunflower 97 %

What we tell the garden centre

Trust it

close-up photos of a single flower

Let staff check

gardens, parks, small flowers; ask for a close-up

Next

more garden photos and an answer for “not one of our three”

Code, slides and video built with an AI assistant. Guesses, decisions and rules: ours.

Thank you